MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY

JAMSHORO

DEPARTMENT OF SOFTWARE ENGINEERING

PROJECT REPORT

HostelMate

University Hostel Updates & Complaints Management System

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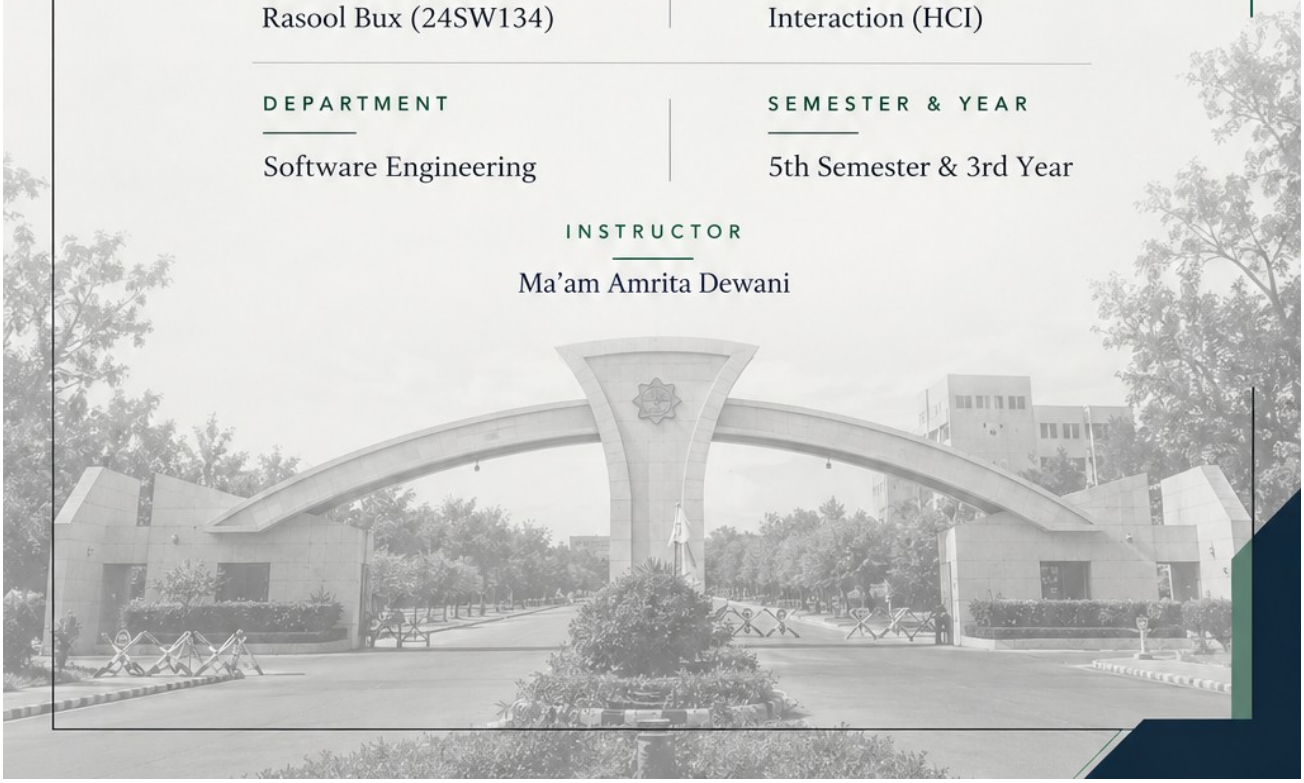
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SEMESTER & YEAR

5th Semester & 3rd Year

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Submission date: _____

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The ten interface screens are presented in the same order in both design sections: Login, Signup, Student Dashboard, Hostel Updates, Update Details, My Complaints, Submit Complaint, Complaint Confirmation, Complaint Status, and Profile.

1 Introduction

HostelMate is a university hostel updates and complaints interface designed for students living in university hostels. It brings announcements and complaint-related tasks into one student-facing experience. Students can read hostel notices, report an issue, review their own complaints, track progress, and view their profile information.

Hostel information may be shared through separate notices or informal communication channels. Students can miss announcements, struggle to find complete details, or remain uncertain about the progress of a reported problem. HostelMate addresses these needs through organized update lists, structured complaint entry, confirmation feedback, and a visible progress view.

The project applies Human-Computer Interaction principles through two versions of the same application. The Good HCI version aims to support clear and efficient task completion. The Bad HCI version intentionally introduces usability weaknesses so their effects can be explained and compared. Each version covers ten corresponding screens.

The scope is an interactive student prototype. Account access, complaint records, and status changes demonstrate interface behavior using sample data; the project does not claim to operate a live hostel service. The evaluation is a qualitative interface analysis, rather than a measured usability study with participants.

2 Target Users

The intended users are university students residing in hostels. They need timely hostel information and a straightforward way to report and follow up on issues such as water leakage, electricity faults, maintenance, and cleanliness.

Students have varying levels of technical experience and may check information quickly on a mobile device. The design therefore uses familiar task names, persistent navigation, clear grouping, and short instructions. The dashboard and complaint list focus on the current student’s own records.

Main tasks: Access the account; find and read announcements; submit a complaint; confirm submission; check complaint progress; and review profile details.

Usability needs: Low input effort, understandable status information, readable content, predictable navigation, and guidance when a form is incomplete.

3 Good HCI Design

Design Overview

The Good HCI design uses a forest-green and cream theme with a repeated card, field, and button system. Dark green identifies primary actions and selected navigation states. Related information appears in separate sections, allowing students to distinguish announcements, complaint records, and account information.

Four labeled bottom-navigation destinations connect the main areas: Dashboard, Updates, Complaints, and Profile. Screen headings and back controls provide local context. The selected destination is highlighted, helping students understand where they are within the application.

The screens follow the student's task sequence. Forms use visible labels and supporting instructions. Complaint status is expressed in text as well as visual indicators. Confirmation information provides a reference number and a route to tracking, making the outcome and next action understandable.

Student Interaction Flow

Account access: Login → Student Dashboard. New students can open Signup and return to Login after the account-created state.

Hostel announcements: Student Dashboard or Updates → Hostel Updates → Update Details → return to the list.

Complaints: My Complaints → Submit Complaint → Complaint Confirmation → Complaint Status → My Complaints.

Account information: Profile → Edit Profile or account-related controls → return to Profile.

Interface Layouts

Figures 1–10 show the student-facing phone view captured from the interactive HTML prototype. External demonstration menus and HCI annotation panels are excluded. Longer screens scroll within the phone; the captures show their initial viewport. Example names and records are demonstration data, rather than the report authors' account details.

3.1.1 Login

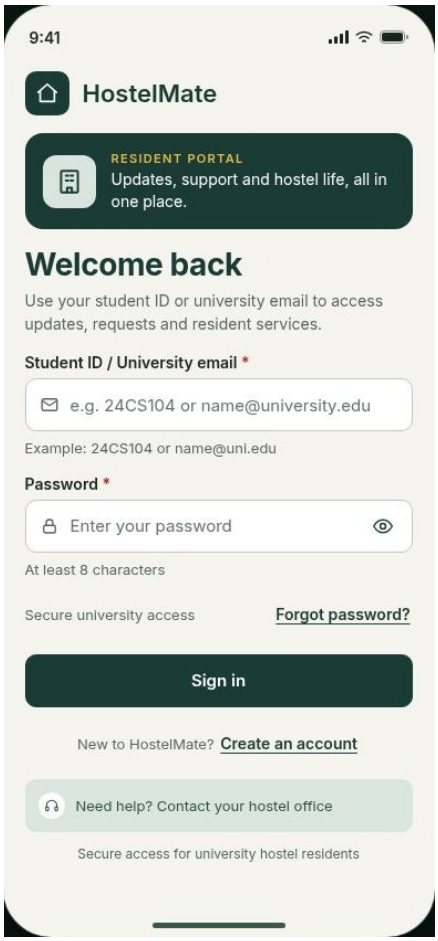


Figure 1 Good HCI Login

Purpose: Provide a clear entry point to the student portal.

Layout and components: A brand header and resident-portal banner introduce the interface. The form contains Student ID / University email and Password fields, a visibility control, a Forgot password link, and a full-width Sign in button.

Navigation: Sign in opens the dashboard after valid demo input. Create an account opens Signup; recovery and hostel-help links provide supporting routes.

Design decisions: Persistent labels and example formats explain the expected input. Password masking supports privacy while the visibility control helps detect typing mistakes. The primary action is visually stronger than supporting links.

HCI principles: Recognition rather than recall, error prevention, familiar language, and clear visual hierarchy.

3.1.2 Signup

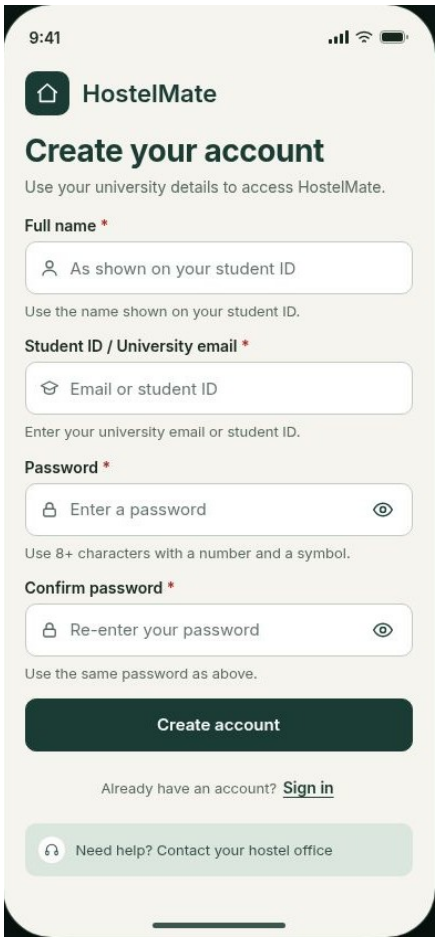


Figure 2 Good HCI Signup

Purpose: Allow students to complete the registration interaction.

Layout and components: Four vertically arranged fields collect Full name, Student ID / University email, Password, and Confirm password. Required markers and instructions appear with the fields. Create account is the main action.

Navigation: Students can return through Sign in. Successful demo registration leads to the account-created state and then Login.

Design decisions: The password instruction appears before submission. Confirmation reduces the chance of a mistyped password. The secondary sign-in route remains identifiable without competing with registration.

HCI principles: Minimal input, consistency, error prevention, and contextual help.

3.1.3 Student Dashboard

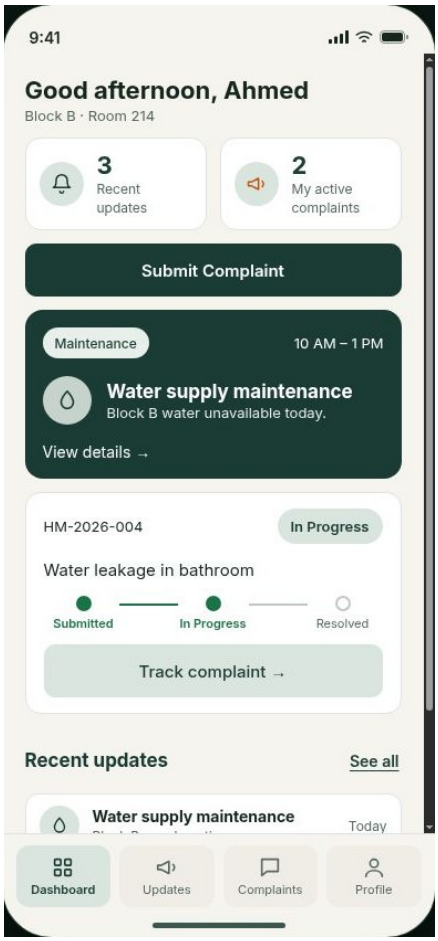


Figure 3 Good HCI Student Dashboard

Purpose: Provide a quick overview and direct access to common hostel tasks.

Layout and components: The greeting and hostel location appear first, followed by update and active-complaint counts, Submit Complaint, a maintenance notice, a personal complaint tracker, and recent announcements.

Navigation: The primary action opens Submit Complaint. Update cards open details; Track complaint opens Complaint Status. Labeled bottom tabs connect the main sections.

Design decisions: An urgent notice receives emphasis while personal complaint progress is visible without opening a separate screen. Cards separate tasks and information; detailed content remains accessible by scrolling.

HCI principles: Visibility of system status, information organization, recognition, and efficient navigation.

3.1.4 Hostel Updates

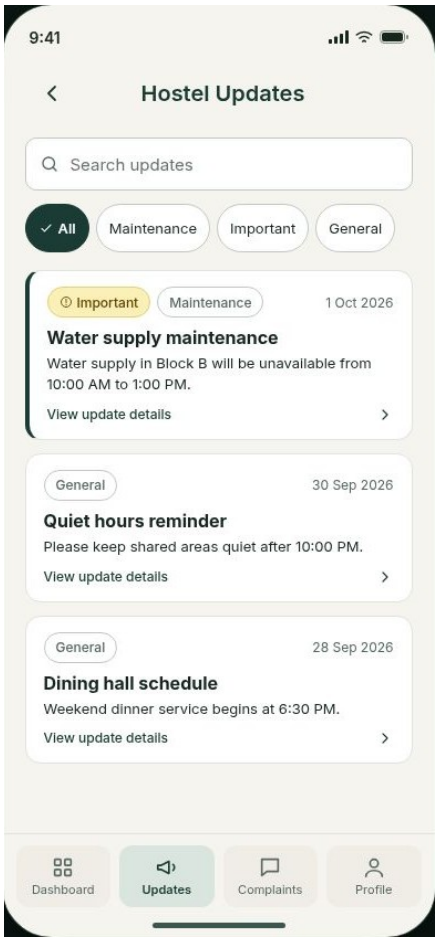


Figure 4 Good HCI Hostel Updates

Purpose: Help students browse and locate hostel announcements.

Layout and components: A search field appears above All, Maintenance, Important, and General filters. Cards contain a title, category information, date, short description, and a detail cue.

Navigation: Selecting an update opens Update Details. Bottom navigation keeps Dashboard, Complaints, and Profile available.

Design decisions: A dark selected filter communicates the current view. Repeated cards support scanning, and search offers a direct way to locate a topic. Dedicated no-result and empty states explain missing content.

HCI principles: Consistency, visibility, recognition rather than recall, and effective information retrieval.

3.1.5 Update Details

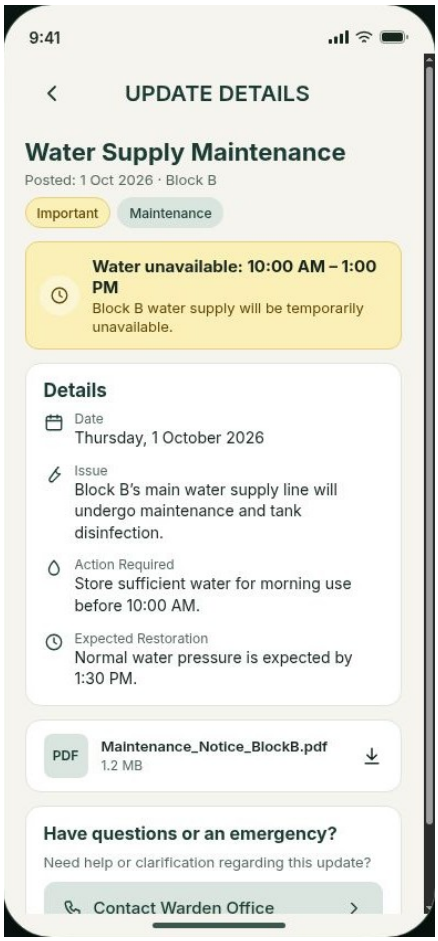


Figure 5 Good HCI Update Details

Purpose: Present complete information about a selected announcement.

Layout and components: The water-maintenance example includes a title, publication information, category tags, a prominent outage summary, and structured Date, Issue, Action Required, and Expected Restoration details. An attachment and contact action follow.

Navigation: The back control returns to the update list. Attachment and Contact Warden Office controls support related actions.

Design decisions: The summary brings timing into focus before the longer details. Separating date, issue, action, and restoration reduces the effort needed to find instructions. Attachment size and type clarify the available resource.

HCI principles: Visual hierarchy, information organization, familiar language, and user control.

3.1.6 My Complaints

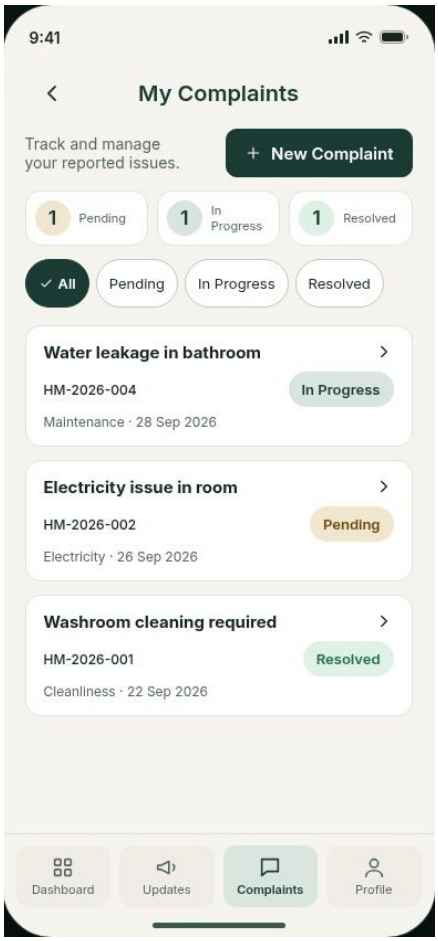


Figure 6 Good HCI My Complaints

Purpose: Allow a student to review their submitted complaints.

Layout and components: The screen presents a New Complaint action, status totals, All and status filters, and complaint cards containing titles, references, dates, and text status badges.

Navigation: New Complaint opens Submit Complaint. Selecting a complaint opens its status view. Bottom navigation links to other main sections.

Design decisions: The example shows one complaint in each status, matching the three displayed records. Text accompanies status color, while consistent entries make titles and references easier to compare.

HCI principles: Visibility of system status, consistency, recognition, and personal relevance.

3.1.7 Submit Complaint

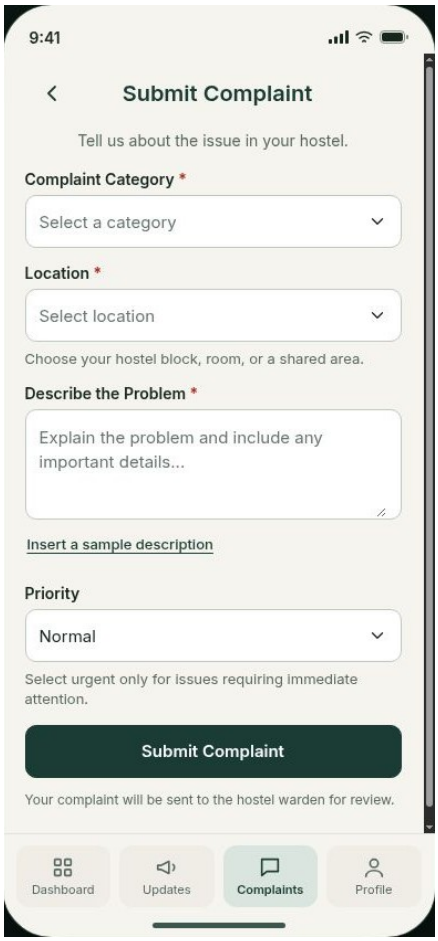


Figure 7 Good HCI Submit Complaint

Purpose: Collect the information needed to describe a hostel issue.

Layout and components: The form contains required Category, Location, and Describe the Problem controls, followed by Priority and a prominent Submit Complaint action. Dropdown cues and short instructions explain the selections.

Navigation: Successful demo submission opens Complaint Confirmation. The back route supports leaving the form; the discard state offers a chance to keep editing.

Design decisions: Selections reduce typing and help structure the report. A multiline description allows sufficient detail. Required-field validation identifies omissions near the affected fields instead of requiring the student to guess.

HCI principles: Minimal input, error prevention, contextual help, and user control.

3.1.8 Complaint Confirmation

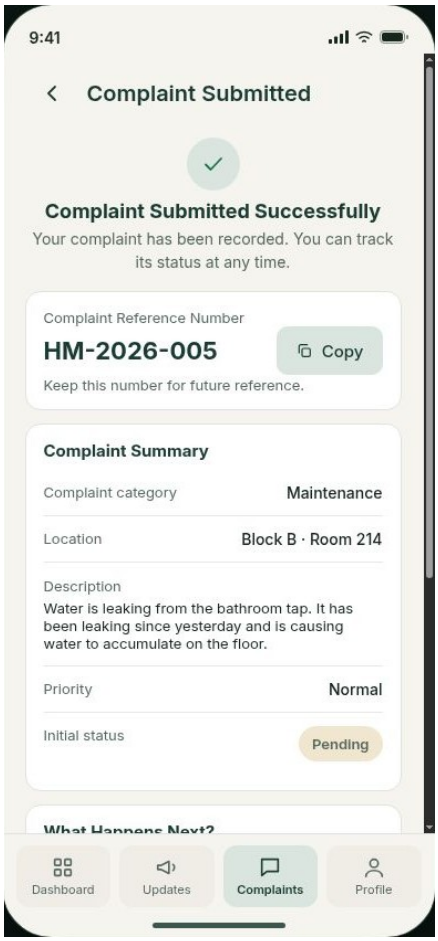


Figure 8 Good HCI Complaint Confirmation

Purpose: Communicate successful submission and the next steps.

Layout and components: The screen presents a success message, a complaint reference with a Copy action, a summary of entered information, an initial status, and a What Happens Next section below.

Navigation: Track Complaint opens tracking; Back to My Complaints returns to the list. These actions are available below the initial viewport.

Design decisions: The reference gives the student a stable identifier. The summary supports checking what was recorded, and the next-step explanation sets expectations about review and progress updates.

HCI principles: Feedback, visibility of system status, recognition, and clear onward navigation.

3.1.9 Complaint Status

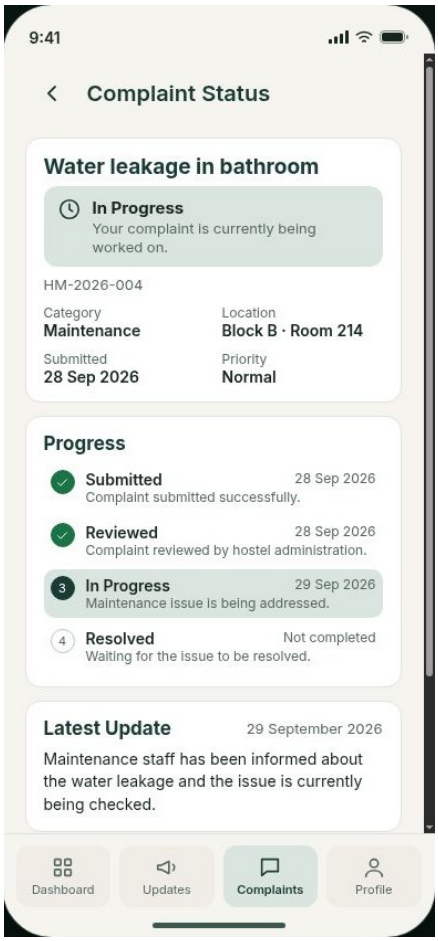


Figure 9 Good HCI Complaint Status

Purpose: Explain the current progress of a selected complaint.

Layout and components: The complaint title and current status lead the page. Reference, category, location, submitted date, and priority are grouped above a Submitted, Reviewed, In Progress, and Resolved timeline and a latest-update section.

Navigation: The back control returns to the complaint list; labeled tabs provide routes to the other main sections.

Design decisions: Completed and current stages have distinct indicators, supported by text explanations. The shown In Progress status agrees with the latest update that the issue is being checked, reducing uncertainty.

HCI principles: Visibility of system status, consistency, understandable feedback, and information organization.

3.1.10 Profile

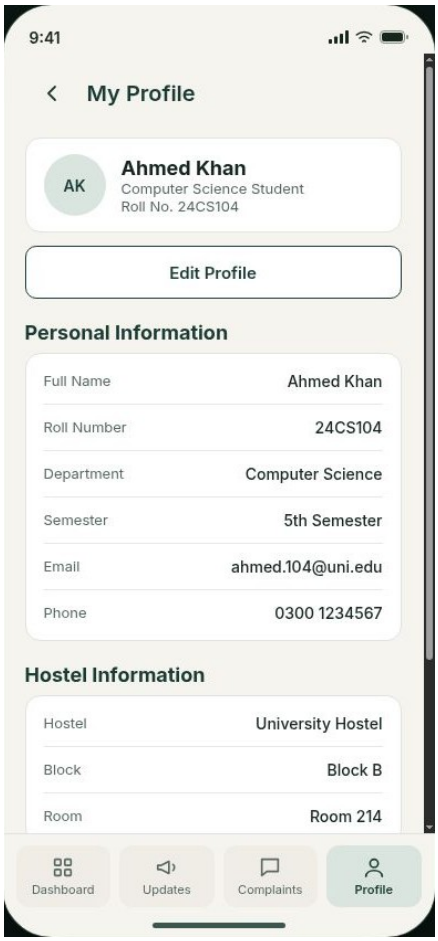


Figure 10 Good HCI Profile

Purpose: Allow a student to review personal and hostel information.

Layout and components: An identity card and Edit Profile action appear above Personal Information and Hostel Information. Label-and-value rows display the student's details; account actions continue below the initial viewport.

Navigation: Edit Profile opens an editable form. Change password and Sign out use separate account controls; navigation returns to the main sections.

Design decisions: Grouping details by topic helps scanning. The edit action is visibly distinct from display-only information. Account actions are separated from routine profile review, with confirmation available for sign-out.

HCI principles: Recognition rather than recall, consistency, user control, and error prevention.

3.2 HCI Principles Applied

Visibility of system status: The dashboard shows active complaints, and the tracking view names the current stage. A selected bottom tab or filter communicates the current location or view. These cues help students understand the interface without repeatedly opening other screens.

Consistency and standards: Repeated green primary buttons, labeled bottom tabs, card layouts, and field styles create a predictable interaction pattern. Students can transfer what they learn on one screen to the next.

Feedback: The confirmation view acknowledges submission and provides a reference number and summary. Validation states identify missing input. Feedback explains both the outcome of an action and what the student should do next.

User control and freedom: Back controls and persistent tabs support movement between tasks. The discard dialog offers Keep editing and Discard complaint, and the sign-out dialog allows cancellation. These options reduce accidental loss of work.

Error prevention: Required markers, password instructions, confirmation input, dropdown choices, and field-specific validation guide entry. The interface presents expectations before submission instead of relying only on an error afterward.

Recognition rather than recall: Visible field labels, named navigation destinations, complaint references, and status text reduce memory demands. Students can identify options rather than remember hidden commands or icon meanings.

Simplicity and user-centered organization: Forms request a small set of task-relevant details. Personal complaint records and grouped hostel information keep the student view focused. Announcements use short summaries with a separate detail screen.

Accessibility considerations: Text labels accompany navigation icons and complaint status colors. Readable hierarchy and distinguishable actions support understanding. These are design considerations; a formal contrast, keyboard, and assistive-technology audit would be needed to establish accessibility compliance.

4 Bad HCI Design

The same student tasks are intentionally presented with poorer interface decisions. Figures 11–20 document the weaknesses and their likely usability effects.

4.1.1 Login

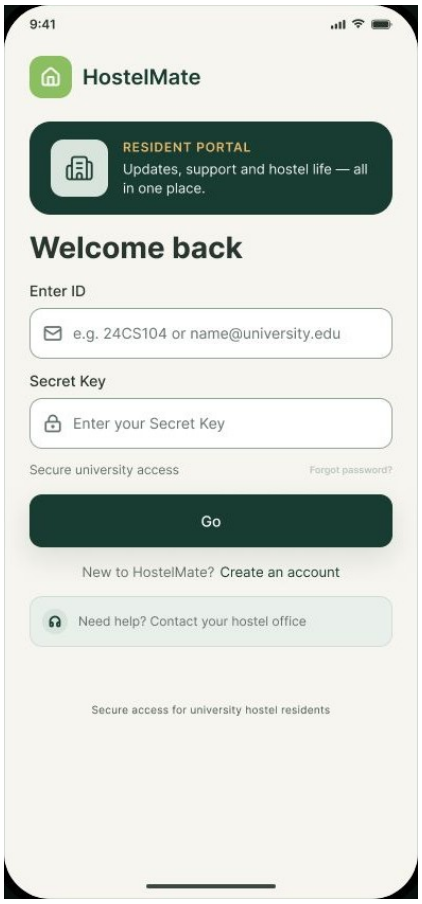


Figure 11 Bad HCI Login

Observed problem: The password is labeled Secret Key, the identifier is called Enter ID, and the main action says Go. The recovery link and supporting text are visually faint.

Principles violated: Match between the system and the real world, recognition rather than recall, and help and documentation.

Usability effect and justification: Students must infer whether Secret Key means a password or another credential and what Go will do. The hidden emphasis of recovery guidance can delay access. Clear Student ID / University email, Password, and Sign in labels in the Good version remove this ambiguity.

4.1.2 Signup

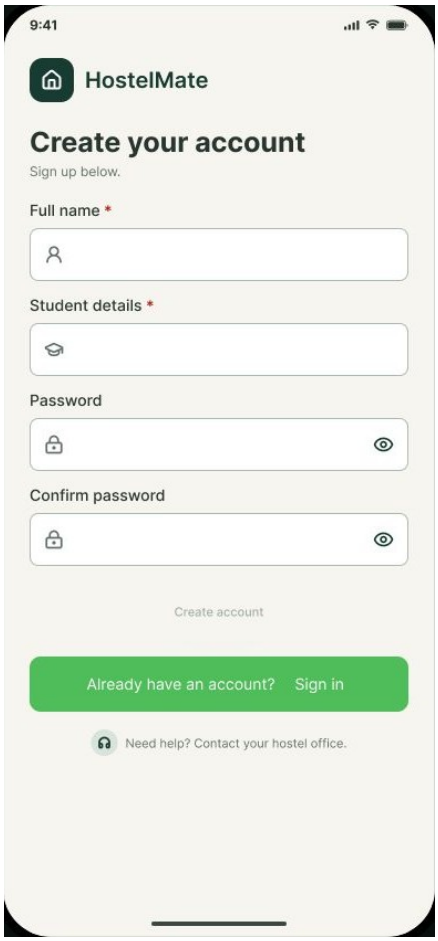


Figure 12 Bad HCI Signup

Observed problem: Student details does not specify the accepted identifier. Create account is faint, while Already have an account? Sign in is a bright full-width button. Password guidance and required markers are inconsistent.

Principles violated: Consistency and standards, error prevention, familiar language, and visual hierarchy.

Usability effect and justification: The returning-user action visually dominates the task of creating an account. New students may select Sign in or overlook required input. The Good version gives registration the primary button and explains each field before submission.

4.1.3 Student Dashboard

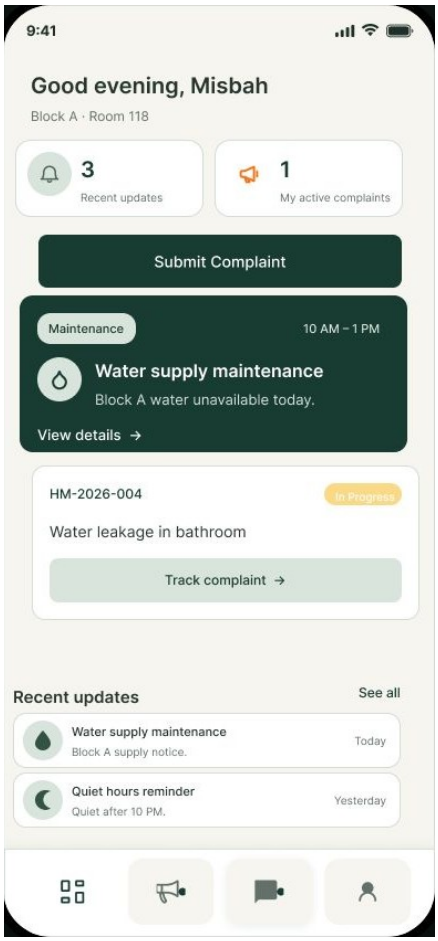


Figure 13 Bad HCI Student Dashboard

Observed problem: Complaint status and tracking controls are pale; secondary descriptions are faint. The bottom bar presents icons without names, and the complaint card offers less visible progress detail.

Principles violated: Visibility of system status, recognition rather than recall, and readable information hierarchy.

Usability effect and justification: Students must inspect small text to understand complaint progress and infer the meaning of each navigation icon. This increases effort during a quick check. The Good dashboard supplies a visible progress indicator and labeled tabs. A short recent-updates preview alone is not treated as an error.

4.1.4 Hostel Updates

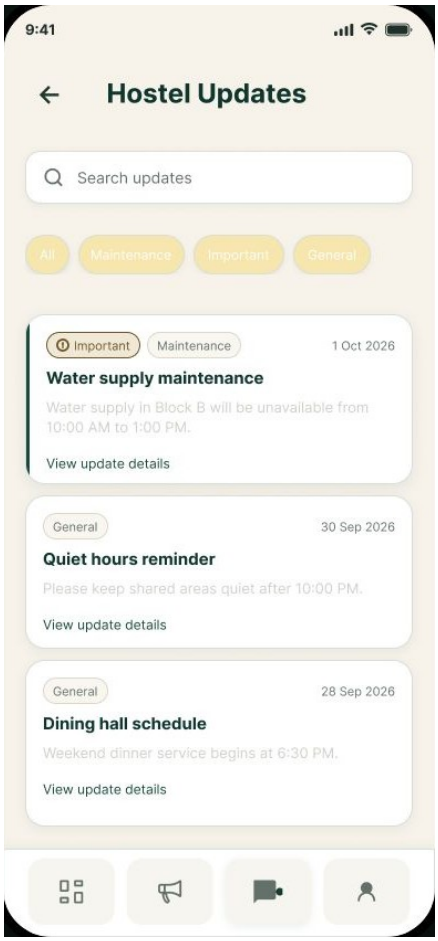


Figure 14 Bad HCI Hostel Updates

Observed problem: Filter chips have pale yellow backgrounds and very light labels, with little distinction between active and inactive filters. Descriptions are faint, and the bottom bar omits destination names.

Principles violated: Visibility of system status, readability, and recognition rather than recall.

Usability effect and justification: Students may struggle to identify a filter or understand which view is active. They can miss the reason a particular set of updates appears. The Good version uses readable labels and a dark selected filter. The weakness concerns legibility and state indication, not the existence of filtering itself.

4.1.5 Update Details

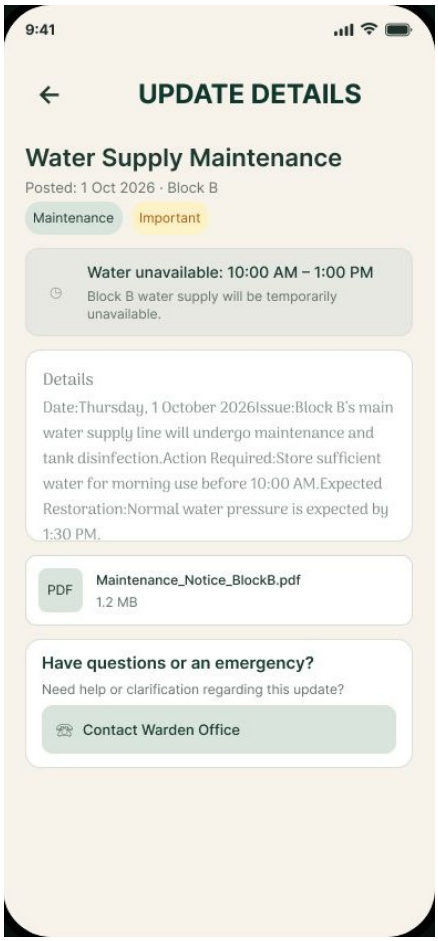


Figure 15 Bad HCI Update Details

Observed problem: The Date, Issue, Action Required, and Expected Restoration information is compressed into a faint paragraph rather than separated into clear detail rows. The main summary also has weaker emphasis.

Principles violated: Information organization, readability, and visual hierarchy.

Usability effect and justification: A student looking for outage timing or an action must search through the paragraph. Important instructions can be missed during a quick read. The Good version provides labeled detail groups and a highlighted timing summary, making the same content easier to locate.

4.1.6 My Complaints

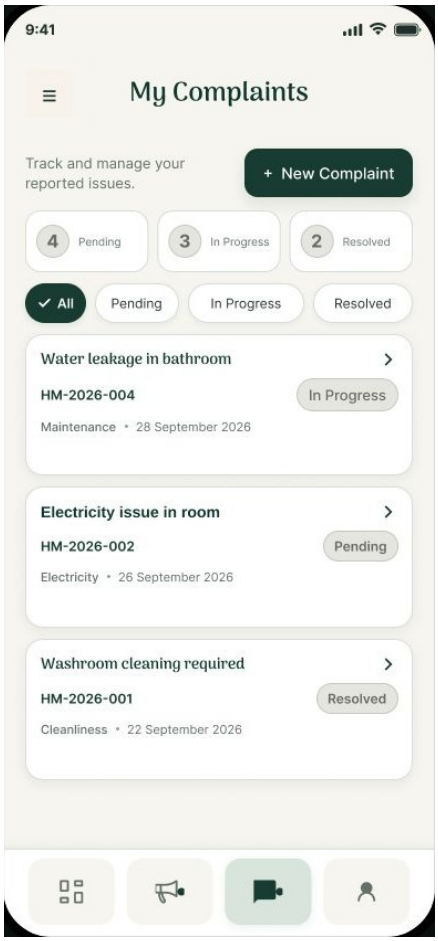


Figure 16 Bad HCI My Complaints

Observed problem: The summary indicates 4 Pending, 3 In Progress, and 2 Resolved complaints, while the All view contains only three records. All status pills look similarly gray, and navigation icons lack labels.

Principles violated: Visibility of system status, consistency, and recognition rather than recall.

Usability effect and justification: Without an explanation for additional records, the totals conflict with the list and reduce confidence in the information. Similar-looking statuses slow scanning, although their text remains meaningful. The Good example uses matching totals and distinct, text-labeled statuses.

4.1.7 Submit Complaint

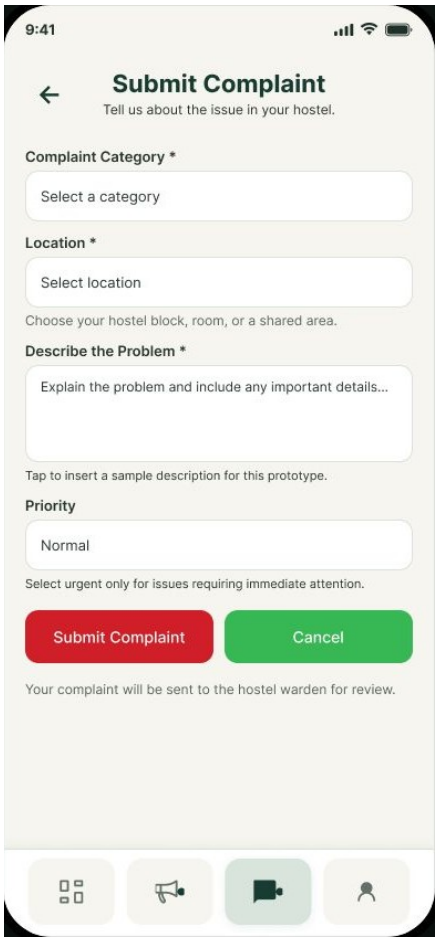


Figure 17 Bad HCI Submit Complaint

Observed problem: Submit Complaint is red while Cancel is bright green, with both actions adjacent and similarly prominent. Selection fields provide weaker dropdown cues, and explanatory text is faint.

Principles violated: Consistency of action emphasis, recognition, error prevention, and readability.

Usability effect and justification: The stronger green Cancel action competes with submission and may attract an unintended tap. Weak selection cues make the form harder to interpret. Color is not the sole reason the design is poor: competing emphasis and unclear control behavior combine to increase avoidable effort.

4.1.8 Complaint Confirmation

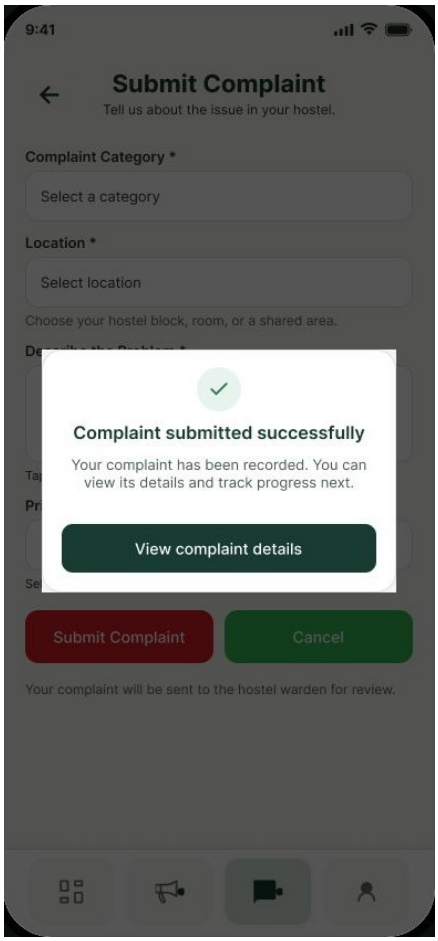


Figure 18 Bad HCI Complaint Confirmation

Observed problem: A generic success overlay contains a single View complaint details button. It does not show the reference number, entered complaint summary, or next-step explanation offered by the Good confirmation view.

Principles violated: Feedback completeness, visibility of system status, and user control.

Usability effect and justification: The overlay confirms an action but gives little information for checking or identifying the complaint. Students must leave it to find details and cannot review the submission immediately. This is limited feedback, rather than an absence of feedback.

4.1.9 Complaint Status

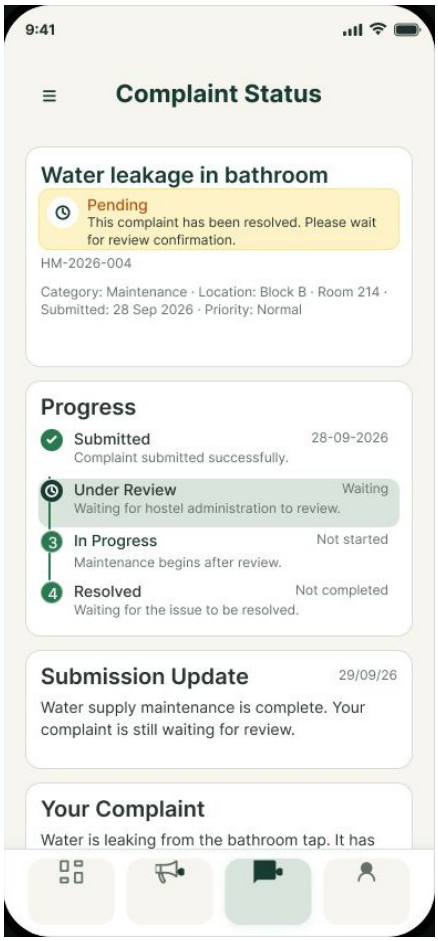


Figure 19 Bad HCI Complaint Status

Observed problem: The badge says Pending while the accompanying message says the complaint has been resolved. The submission update says maintenance is complete while review remains pending. Metadata is compressed into small text.

Principles violated: Visibility of system status, consistency, and information organization.

Usability effect and justification: The conflicting statements prevent a reliable understanding of progress: students cannot tell whether to wait or consider the issue closed. This can cause unnecessary follow-up and reduce trust. The Good screen aligns its current status, progress stage, and latest update.

4.1.10 Profile

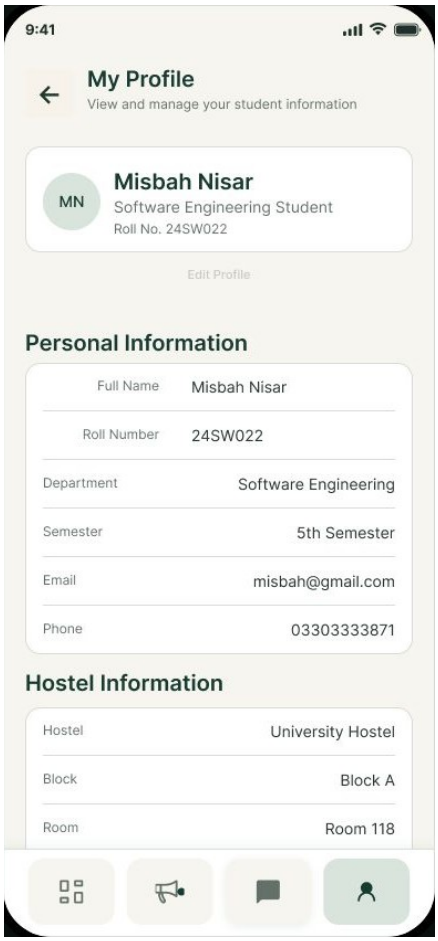


Figure 20 Bad HCI Profile

Observed problem: Edit Profile is faint and does not look like a prominent action. Personal details are less consistently aligned, and bottom-navigation icons have no text labels. Hostel details continue beyond the initial visible region.

Principles violated: Recognition rather than recall, consistency, and readable information organization.

Usability effect and justification: Students may overlook editing or need more effort to scan label-and-value pairs. Unlabeled navigation adds guesswork. Scrolling is not inherently a problem; the weakness is making available information and actions difficult to discover. The Good version uses a clear edit button and grouped rows.

5 Good and Bad HCI Comparison

The comparison below links each design difference to a student task. The effects are qualitative expectations from interface analysis; no completion-time, error-rate, or satisfaction measurements are claimed.

HCI aspect	Good design	Bad design
Language	Student ID / University email, Password, and Sign in identify the task.	Enter ID, Secret Key, and Go require interpretation.
Primary actions	Create account is the strongest signup action.	The sign-in route dominates the faint account-creation action.
Navigation	Bottom icons have destination labels and a selected state.	Icons omit names, requiring users to infer destinations.
Update filtering	Readable chips distinguish the active filter.	Pale chips make labels and selected state difficult to identify.
Information layout	Update details separate timing, issue, action, and restoration.	A faint paragraph makes the same facts harder to scan.
Complaint totals	One Pending, one In Progress, and one Resolved record agree with the list.	The summary counts exceed the records in the All view without explanation.
Form actions	A prominent submit action supports the intended task.	Red Submit and bright green Cancel compete side by side.
Confirmation	Reference, summary, and next steps support verification and tracking.	A generic success overlay omits the reference and summary.
Status accuracy	Current status, timeline, and latest update agree.	Pending and resolved messages conflict on the same complaint.
Profile and readability	A visible edit button and aligned groups support review.	A faint edit action and weaker alignment reduce discoverability.

6 Design Tools and Prototype

Figma: Used to design screen layouts, interface components, typography, spacing, colors, and corresponding Good and Bad design states. The two versions maintain a common application scope so their usability differences can be compared.

HTML prototypes: Used to demonstrate navigation and interactive entry in a browser. The report figures show the student phone view, while surrounding demonstration controls are excluded from the screenshots.

Good prototype: https://hostelmatehci.netlify.app/

Bad prototype: https://hostelmate-bad-hci.netlify.app/

The prototypes demonstrate account access, announcement browsing, complaint entry, confirmation, tracking, and profile interaction with sample content. They support an HCI demonstration without requiring a production backend or real authentication.

7 Conclusion

HostelMate applies HCI principles to a practical student need: finding hostel information and managing personal complaints. Its ten connected screens cover the complete student interaction sequence from account access to announcements, complaint submission, tracking, and profile review.

The Good design supports understanding through familiar labels, visible primary actions, structured content, consistent navigation, and meaningful confirmation. The Bad design demonstrates how a visually similar application can still become difficult through ambiguous wording, weak action emphasis, unreadable filters, unexplained totals, and contradictory progress information.

The comparison shows that usability depends on both presentation and the accuracy of the information communicated. A confirmation must help students identify what happened, and a status view must tell a coherent story about progress. The project demonstrates these differences through corresponding screens and interactive prototypes.

Further evaluation could observe students completing the same tasks in both versions and compare errors, task completion, and feedback. Such testing would provide evidence beyond the qualitative analysis presented in this report.

8 References

1. HCI Project Submission Guidelines. Course-provided project instructions, Human-Computer Interaction, MUET Jamshoro.

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